CO4-AT3-Q1:

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TSP vs Floyd's Algorithm (Path Optimization)
Problem Understanding

Path optimization problems find the shortest routes between multiple locations. TSP finds the minimum cost tour visiting all cities once, while Floyd's Algorithm finds shortest paths between every pair of nodes.

Algorithms Used

Travelling Salesman Problem (TSP)

- Finds minimum route covering all cities.
- Returns to starting city.
- Uses optimization approach.

Floyd's Algorithm

- Computes shortest path between all pairs of vertices.
- Uses Dynamic Programming.

Comparison

Feature	TSP	Floyd Algorithm
Purpose	Single best tour	All shortest paths
Technique	Optimization	Dynamic Programming
Complexity	$O(n!)$	$O(n^3)$
Scalability	Limited	Better for large graphs

Result Analysis

Floyd's Algorithm efficiently calculates multiple shortest paths, while TSP provides the optimal travel route but becomes expensive for large numbers of cities.

Conclusion

For small delivery networks TSP is useful, but for large-scale path analysis Floyd's Algorithm provides better scalability and performance.